

DATA ANALYTICS WITH POWER BI PROJECT REPORT

TITLE: SPOTIFY GLOBAL TOP 50 DASHBOARD

Submitted by

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CERTIFICATE

This is to certify that **Yogesh Yadav**, Registration Number **12312204**, a student of the **B.Tech Computer Science and Engineering** program, has successfully completed the project titled "**Spotify Global Top 50 Dashboard**" under my guidance and supervision.

The project is a bona fide work carried out by him as a part of the course **INT374 – Data Analytics with Power BI**. To the best of my knowledge, the matter embodied in this project report has not been submitted to any other university or institute for the award of any degree or diploma. The project demonstrates a comprehensive understanding of data analytics principles, data preprocessing, data modeling using DAX, and dashboard visualization techniques applied to real-world music streaming datasets. The work presents an in-depth analysis of global music consumption trends, artist performance metrics, and algorithmic popularity drivers using the provided Spotify datasets and customized business intelligence tools.

Date: December 24, 2025

Place: Phagwara, Punjab

(Signature of Faculty Mentor)

Name: Savleen Kaur

DECLARATION

I, **Yogesh Yadav**, bearing Registration Number **12312204**, hereby declare that the project report titled "**Spotify Global Top 50 Dashboard**" submitted by me to the **Lovely School of Computer Science and Engineering**, is a record of original work done by me under the guidance of **Ms. Savleen Kaur**.

I further declare that this work is original and has not been submitted, in part or full, for any other degree or diploma to this or any other university. All sources of data, including the *spotify-top-50-world.csv* dataset, and theoretical concepts used in this report have been duly acknowledged and cited. The analysis, visualizations, and insights presented herein are the result of my own efforts in applying Power BI tools to the provided data.

Date: December 24, 2025

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ACKNOWLEDGEMENT

The successful completion of this project, "**Spotify Global Top 50 Dashboard**," would not have been possible without the support and guidance of several individuals who contributed to the academic and technical rigor of this undertaking.

First and foremost, I would like to express my sincere gratitude to my faculty mentor, **Ms. Savleen Kaur**, for her invaluable guidance, constant encouragement, and constructive feedback throughout the duration of this course **INT374**. Her deep knowledge of Data Analytics, DAX implementation, and Power BI visualization best practices has been instrumental in shaping the direction of this project and refining the complexity of the data model.

I would also like to thank the **Lovely School of Computer Science and Engineering** for providing the necessary infrastructure, computer labs, and software resources required to carry out this project effectively. The curriculum design and the practical labs provided a robust foundation for understanding the complexities of business intelligence tools.

Finally, I acknowledge the open-source community and data contributors on platforms like Kaggle and Spotify Developers, whose datasets served as the backbone of this analysis. The availability of rich, granular data on global music trends allowed for a nuanced exploration of the streaming landscape, enabling the simulation of a real-world music industry analytics

scenario.

ABSTRACT

The global music industry has undergone a seismic shift from physical sales to digital streaming, generating a deluge of data that captures real-time cultural consumption. This project, "**Spotify Global Top 50 Dashboard**," leverages **Microsoft Power BI** to engineer a comprehensive business intelligence solution that analyzes the volatility, artist dominance, and content strategies within the Spotify Global Top 50 charts. Utilizing a high-frequency dataset (spotify-top-50-world.csv) covering the period from mid-2023 through 2024, the report documents the end-to-end development of an analytical dashboard designed for industry stakeholders.

The primary objective was to deconstruct the "anatomy of a hit" by correlating metadata—such as song duration, explicit content markers, and release types (single vs. album)—with performance metrics like Popularity scores and daily chart positions. The project employed advanced **Data Analysis Expressions (DAX)** to construct a dynamic data model, calculating complex measures such as *Position 1 Hits per Artist*, *Average Popularity by Explicit Status*, and *Stream Velocity*.

Key findings from the analysis indicate a distinct consolidation of power among "superstar" artists, with Taylor Swift and Bad Bunny commanding disproportionate chart real estate. The data reveals a significant trend toward shorter song durations (averaging roughly 3 minutes and 28 seconds in the top tier), driven by streaming economics that incentivize replayability. Furthermore, the analysis confirms the dominance of the "Single" format in gaining initial chart traction, although "Album" releases from major artists drive sustained engagement. The dashboard effectively visualizes these trends through interactive elements, including a Home landing page, an Artist-centric deep dive, and a granular Song analysis view. This report details the technical implementation of the ETL pipeline, the logic behind the DAX measures, and the strategic insights derived, offering a microcosm of the current state of the global music streaming economy.¹

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1. SOURCE OF DATASET

The integrity and depth of any data analytics project are fundamentally tethered to the quality of its source material. For this dashboard, the primary dataset utilized is the **Spotify Global Top 50 World Dataset** (spotify-top-50-world.csv), a structured repository of daily chart rankings that serves as a proxy for global musical zeitgeist.

1.1 Dataset Origin and Context

The data is derived from Spotify's public API and aggregated repositories (such as Kaggle), which track the daily movement of the "Top 50 - Global" playlist. This chart is one of the most influential metrics in the modern music industry, often dictating marketing spend, tour routing, and radio play. Unlike viral charts which measure velocity (rate of change), the Top 50 measures absolute consumption magnitude, combining paid subscription streams and ad-supported streams.² The dataset spans a critical transition period in the industry (2023-2024), capturing the rise of regional Mexican genres (e.g., Peso Pluma) and the global dominance of K-Pop alongside traditional Western Pop.¹

1.2 Data Dictionary and Attributes

The raw CSV file contains a rich matrix of categorical and numerical data points. Understanding these attributes was essential for the subsequent data modeling phase:

Column Name	Data Type	Description	Relevance to
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			Analysis
date	Date	The specific calendar date of the chart entry.	Crucial for time-series analysis and identifying seasonality.
position	Integer	The rank of the song (1-50).	The primary metric of success; used to calculate "Peak Position".
song	Text	The title of the track.	Used for distinct counts and granular drill-downs.
artist	Text	The performing artist(s).	Key for aggregating "Artist Dominance" and "Hits per Artist".
popularity	Integer	A score (0-100) calculated by Spotify.	Indicates current trendiness relative to the rest of the catalog. ²
duration_ms	Integer	Track length in milliseconds.	Converted to minutes to analyze the "short song" trend. ⁶
album_type	Text	"Album", "Single", "Compilation".	Essential for the "Singles vs. Albums" strategy analysis.
total_tracks	Integer	Track count of the parent album.	Correlated with "Album" type; helps identify EP

			releases.
release_date	Date	The official drop date of the track.	Used to calculate "Days Since Release" and shelf-life.
is_explicit	Boolean	TRUE/FALSE indicator.	Used to analyze the prevalence of explicit content in top tiers. ⁷
album_cover_url	URL	Web link to artwork.	Used in the dashboard for visual context (Image display).

1.3 Volume and Granularity

The dataset captures daily snapshots. This high-frequency granularity allows for the observation of specific phenomena, such as "the weekend drop" (where streams might dip or spike depending on the genre) and the immediate impact of an album release on the charts. By analyzing thousands of rows representing daily positions, the dashboard can smooth out noise and identify genuine, sustained trends rather than just momentary viral spikes. The presence of metadata like ISRC (International Standard Recording Code) in related datasets ⁸ ensures uniqueness, though the primary analysis here relies on the song and artist composite keys to track performance.

2. INTRODUCTION

The **Spotify Global Top 50 Dashboard** represents a comprehensive application of Business Intelligence (BI) principles to the domain of digital music streaming. In an era defined by the "Attention Economy," music consumption has shifted from a transactional model (buying a CD) to an access-based model (streaming). This shift has generated exabytes of data, transforming the role of a music executive or analyst into one that requires deep data literacy. This project utilizes **Microsoft Power BI** to decode the signals hidden within the Spotify charts.

2.1 The Strategic Importance of Streaming Analytics

The music industry is currently characterized by hyper-saturation. With over 100,000 new

tracks uploaded to Digital Service Providers (DSPs) daily ⁹, the competition for a spot in the Global Top 50 is fiercer than ever. Achieving a position in this chart is not merely a vanity metric; it correlates directly with revenue, global touring opportunities, and brand partnerships. Therefore, understanding the *characteristics* of the songs that make it to this elite list is of paramount importance. Does the market favor shorter songs? Are "Singles" more effective than "Albums" for breaking new artists? How does explicit content impact playability? This dashboard seeks to answer these questions through rigorous data visualization.

2.2 Project Scope and Utility

This report outlines the creation of a dashboard that serves multiple personas:

- **The A&R Manager:** Who looks for trending artists and genres (e.g., the explosion of Latin music represented by Bad Bunny and Peso Pluma).
- **The Marketing Director:** Who needs to understand the seasonality of popularity (e.g., when does "All I Want for Christmas Is You" enter the chart?).
- **The Artist Manager:** Who analyzes the shelf-life of a track to determine when to release a remix or a follow-up single.

The dashboard integrates disparate data points—ranking, audio features, and metadata—into a cohesive narrative. By moving beyond static spreadsheets to interactive visuals, the project enables "drill-down" capabilities, allowing a user to start with a global view and zoom into the performance of a specific track like *Espresso* or *Die With A Smile*.¹⁰

2.3 Technological Framework

The project is built upon the Microsoft Power BI ecosystem, leveraging its three core pillars:

1. **Power Query (M Language):** For robust Extract, Transform, and Load (ETL) processes to clean the raw CSV data.
2. **Power Pivot (DAX):** For creating a semantic model that defines relationships and calculates dynamic measures.
3. **Power View:** For the front-end visualization layer that presents the data in an accessible, aesthetic format.

This report serves as a definitive documentation of the project, detailing the methodology from raw data ingestion to the final generation of business insights.

3. DATA PREPROCESSING & ETL

The raw dataset, while structured, required a significant amount of cleaning and transformation to be analysis-ready. The **ETL (Extract, Transform, Load)** process was executed within the Power Query Editor to ensure data integrity and optimize the

performance of the Power BI model.

3.1 Data Profiling and Quality Assessment

Initial profiling of spotify-top-50-world.csv revealed several data quality conditions that needed addressing:

- **Data Type Mismatches:** The date and release_date columns were initially interpreted as text strings. This prevents any time-series analysis or date-based filtering.
- **Boolean Text:** The is_explicit column contained text values "TRUE" and "FALSE". While usable, converting these to binary integers (1/0) or strictly typed logicals allows for easier aggregation in DAX (e.g., summing the column effectively counts the explicit tracks).
- **Unit Inefficiency:** The duration_ms column provided track length in milliseconds (e.g., 234352). This is not a human-readable format for a dashboard and required conversion to minutes.
- **Dimensionality:** The dataset is a "flat file," meaning specific artist or album details are repeated for every day a song charts. This increases file size and required careful handling during measure creation to avoid over-counting distinct entities.

3.2 Transformation Steps (Power Query)

The following specific transformations were applied to the query Top-50-world:

1. **Header Promotion:** The first row of the CSV was promoted to headers to correctly identify fields like position, song, and popularity.
2. **Date Standardization:**
 - The date column was cast to the **Date** data type. Locale settings were adjusted to ensure the dd-mm-yyyy format was parsed correctly to prevent date inversion errors.
 - The release_date column contained mixed granularity (some tracks had full dates, others just years). A conditional transformation logic was used to standardize year-only entries (e.g., "2023") to "01/01/2023" to maintain a consistent Date data type across the column.
3. **Duration Calculation:**
 - A Custom Column named Duration_Mins was created using the M formula: `Table.AddColumn("#PreviousStep", "Duration_Mins", each [duration_ms] / 60000, type number)`
 - This converts the millisecond value into decimal minutes (e.g., 3.5 minutes), which is far more interpretable for the end-user.
4. **Explicit Content Normalization:**
 - The is_explicit column was duplicated and transformed. One version was kept as a clear "Explicit/Non-Explicit" text category for visual legends, while the other was verified as a logical TRUE/FALSE for DAX calculations like ````.
5. **Text Cleaning:**
 - The album_type column was cleaned using the "Capitalize Each Word" transform to ensure consistency (e.g., converting "single" to "Single") to prevent fragmentation in

- o slicers and donut charts.
- o Artist names were trimmed of leading/trailing whitespace to ensure distinct counts were accurate.

3.3 Data Loading and Storage

Post-transformation, the data was loaded into the VertiPaq engine. The resulting table provided a clean, typed foundation for the star schema model (or in this case, a highly optimized single-table model suitable for the specific dataset scope). The ETL process ensured that downstream DAX measures would perform efficiently without needing to handle type conversions at query time.

4. OBJECTIVES DESCRIPTION

The dashboard was conceptualized to answer specific high-value business questions relevant to music industry stakeholders. These objectives guided the design of the visualizations and the complexity of the DAX measures.

4.1 Primary Analytical Objectives

1. **Global Trend Monitoring:** To visualize the daily ebb and flow of track popularity. The dashboard aims to answer: *Which songs are currently trending, and how volatile is the top of the chart?* This requires time-series visualization capabilities.
2. **Artist Dominance and Market Share:** To quantify the "power law" in music streaming. The objective is to identify which artists (e.g., Taylor Swift, Bad Bunny) occupy the most chart real estate. We aim to measure this not just by stream count, but by the number of unique tracks appearing in the Top 50 simultaneously.¹¹
3. **Release Strategy Evaluation (Singles vs. Albums):** To determine the most effective format for chart success. The dashboard analyzes the album_type distribution to see if the market favors the high-frequency release of "Singles" or the cohesive drops of "Albums." This is critical for labels deciding on release strategies.¹²
4. **Audio Feature Correlation:** To investigate the relationship between song attributes and success. Specifically, *is the average song duration decreasing?* Industry research suggests shorter songs perform better on streaming due to replay incentives.⁶ This dashboard tests that hypothesis against the actual 2023-2024 data.
5. **Content Analysis (Explicit vs. Clean):** To assess the prevalence of explicit content in the global mainstream. Does an "Explicit" rating hinder a song's ability to reach the Top 50, or is it a non-factor in the streaming era?⁷

4.2 KPI Definition

To track these objectives, specific Key Performance Indicators (KPIs) were defined for the

dashboard:

- **Avg Popularity:** The mean popularity score, serving as a baseline for "hit" status.
- **Distinct Songs & Artists:** Measures of chart diversity.
- **Pct Explicit Songs:** The ratio of explicit to total tracks, monitoring content trends.
- **Position 1 Hits:** A count of how many tracks achieved the coveted #1 spot.

5. DATA MODELING & DAX IMPLEMENTATION

The analytical core of the dashboard is built on **Data Analysis Expressions (DAX)**. While the raw data provides the *facts*, DAX provides the *insights* by allowing for dynamic aggregation based on the user's interaction with the report (e.g., filtering by Artist or Date).

5.1 Measures and Logic

The following measures were implemented, following the specific logic required to analyze the Spotify dataset.

1. Volume and Diversity Measures:

To understand the scale of the data, we first define basic counting measures.

- **Total Songs:** `MEASURE 'Top-50-world' = COUNTROWS('Top-50-world')`
 - *Logic:* Counts every row in the current context. This represents the total "chart appearances."
- **Distinct Songs:** `MEASURE 'Top-50-world' = DISTINCTCOUNT('Top-50-world'[song])`
 - *Logic:* Counts unique song titles. If "Flowers" charts for 30 days, it counts as 1 distinct song but 30 total song appearances.
- **Distinct Artists:** `MEASURE 'Top-50-world' = DISTINCTCOUNT('Top-50-world'[artist])`
 - *Logic:* Identifies the number of unique primary artists in the selection.

2. Popularity and Performance Measures:

These measures quantify success.

- **Avg Popularity:** `MEASURE 'Top-50-world'[Avg Popularity] = AVERAGE('Top-50-world'[popularity])`
 - *Logic:* Returns the mean popularity score. This is dynamic; if a user filters for "Taylor Swift," it shows her specific average popularity.
- **Max Popularity:** `MEASURE 'Top-50-world'[Max Popularity] = MAX('Top-50-world'[popularity])`
- **Position 1 Hits per Artist:**
Code snippet
`MEASURE 'Top-50-world'[Position1 Hits per Artist] =
CALCULATE(COUNTROWS('Top-50-world'), 'Top-50-world'[position] = 1)`
 - *Logic:* Uses `CALCULATE` to override the context and filter only for rows where the

position is 1. When dropped into a visual with "Artist," it counts how many days that artist spent at #1.

3. Content Analysis Measures (Explicit vs. Duration):

These measures help answer the specific objectives regarding song attributes.

- **Explicit Songs Count:**

Code snippet

```
MEASURE 'Top-50-world' =  
CALCULATE(COUNTROWS('Top-50-world'), 'Top-50-world'[is_explicit] = TRUE())
```

- **Percentage of Explicit Songs:**

Code snippet

```
MEASURE 'Top-50-world' =  
DIVIDE(, 0)
```

- *Logic:* Provides a normalized ratio (e.g., 0.35 or 35%). Using DIVIDE handles potential divide-by-zero errors safely.

- **Average Duration (Minutes):**

Code snippet

```
MEASURE 'Top-50-world' =  
AVERAGE('Top-50-world'[duration_ms]) / 60000
```

- *Logic:* Aggregates the millisecond duration and divides by 60,000 to return the value in minutes.

4. Segmentation Measures (Singles vs. Albums):

- **Singles Count:**

Code snippet

```
MEASURE 'Top-50-world' =  
CALCULATE(COUNTROWS('Top-50-world'), 'Top-50-world'[album_type] = "single")
```

- **Albums Count:**

Code snippet

```
MEASURE 'Top-50-world'[Albums Count] =  
CALCULATE(COUNTROWS('Top-50-world'), 'Top-50-world'[album_type] = "album")
```

5.2 Context and Evaluation

The EVALUATE SUMMARIZECOLUMNS logic (provided in the setup) demonstrates how these measures interact. For example, by summarizing and, the model allows us to compare the intensity of 2023 versus 2024. The DAX implementation ensures that all KPIs—from "89.62 Avg Popularity" to "342 Count of Artist"—remain consistent across different views of the

dashboard (Home, Overview, Artist, Songs).

6. DASHBOARD DESIGN & VISUALIZATION

The dashboard visualization strategy focuses on clarity, aesthetic alignment with the subject matter (using Spotify's signature black and green palette), and high interactivity. The report generated four distinct views (Home, Overview, Artists, Songs) to allow users to navigate from high-level KPIs to granular details.

6.1 Dashboard Structure and Navigation

- **Visual Style:** A dark-mode theme was selected to mimic the Spotify app interface, reducing eye strain and enhancing the vibrant green data accents.
- **Navigation:** Buttons for "Home," "Overview," "Artists," and "Songs" were placed consistently at the top right (as seen in Image 1 and 2), utilizing Power BI's page navigation actions to create an "app-like" experience.

6.2 View 1: The Overview Dashboard (Image 2 Analysis)

The "Overview" page serves as the executive summary.

- **KPI Cards:** Key metrics are displayed prominently: **789 Distinct Songs**, **342 Artists**, and an **Avg Popularity of 89.62**. This immediately tells the stakeholder that the Top 50 is a high-churn environment (789 unique songs appearing suggests high turnover).
- **Donut Charts (Segmentation):**
 - *Distinct Songs by Album Type:* This visual breaks down the catalog into "Single" vs. "Album". The visual shows roughly **269 Singles** vs **562 Album tracks**, providing immediate insight into the volume of album bombs (where an artist releases an album and all tracks chart).
 - *Explicit vs. Non-Explicit:* A donut chart visualizes the content rating. The split (approx. **11K Explicit** vs **17K Non-Explicit**) suggests that while clean content is the majority, explicit content holds a massive 39-40% share, confirming it is not a barrier to mainstream success.
- **Trend Analysis (Line Chart):** "Avg Popularity by Month" plots the trajectory of popularity over the year. It shows a stable trend line hovering between 88-92, with dips and spikes corresponding to major market shifts or holidays (e.g., December spikes for classics like Mariah Carey).
- **Bar Chart (Distinct Songs by Month):** This histogram shows the volume of unique songs entering the chart per month. A higher bar indicates a volatile month with many new releases (e.g., July/August), while lower bars indicate a stagnant chart dominated by long-running hits.

6.3 View 2: Artist Insights (Image 3 Analysis)

The "Artists" page allows for deep dives into specific performers.

- **Top Artists Bar Chart:** This horizontal bar chart ranks artists by "Position 1 Hits" or "Distinct Songs". **Taylor Swift** dominates this view with **85 distinct songs**, followed by Travis Scott (30) and Drake (27). This visual confirms the "superstar economy" where a single artist captures a massive share of the Top 50.
- **Detailed Matrix:** A table view lists specific tracks, release dates, and popularity scores. For example, it lists "Slut! (Taylor's Version)" with an average popularity of 85.76. This allows A&R managers to see exactly *which* tracks from an album are performing best.

6.4 View 3: Song Analysis (Image 4 Analysis)

The "Songs" page focuses on the tracks themselves.

- **Comparisons:** "Songs by Popularity" compares tracks like "I Wanna Be Yours" (51K popularity sum) against "Cruel Summer" (50K).
- **Hits per Artist:** A visual ranking showing **Lady Gaga** and **Jung Kook** with high "Position 1 Hits," indicating their ability to debut at the top of the charts.
- **Detailed Table:** A grid listing artists alongside metrics like "Avg Tracks per Album" and "Distinct Songs." This table reveals that while **Taylor Swift** has 83 distinct songs, an artist like **Ariana Grande** might have fewer songs but a higher average popularity per track, suggesting a "quality over quantity" or "single-focused" strategy.

7. ANALYSIS & INSIGHTS

By synthesizing the Power BI visualizations with the underlying dataset and broader industry context, we derive several critical insights regarding the state of global music in 2024.

7.1 The "Superstar" Consolidation

The analysis of the `` and [Position 1 Hits] measures reveals a stark consolidation of power.

- **Data Point:** Taylor Swift alone accounts for **85 distinct songs** in the dataset [Image 3]. This is not merely popularity; it is market saturation. When Swift releases an album like *The Tortured Poets Department*, nearly every track enters the Top 50.¹⁰
- **Implication:** The "Album Bomb" is a valid and dominant strategy for superstars. Unlike the singles-driven market for developing artists, established giants like Swift, Bad Bunny, and The Weeknd leverage their massive catalogs to occupy 20-30% of the Top 50 chart simultaneously upon release. This makes it incredibly difficult for new artists to break through during these "blockade" periods.

7.2 The Strategic Dominance of "Singles"

While superstars release albums, the broader market relies on singles.

- **Data Point:** The dashboard's "Distinct Songs by Album Type" visual shows that **Singles** make up a substantial portion of the chart entries. Furthermore, viral hits like "Espresso" by Sabrina Carpenter and "Gata Only" ¹¹ entered as singles.
- **Insight:** The algorithm favors consistency. Releasing singles every 4-6 weeks (drip-feeding) keeps an artist's "Monthly Listeners" metric high and triggers the Spotify algorithm more frequently than a single album drop every two years.¹² The data suggests that for non-superstars, the Single is the primary vehicle for entering the Top 50.

7.3 The Shrinking Song Duration

The `` measure highlights a significant industry trend: songs are getting shorter.

- **Data Point:** The average duration calculated in the dashboard is roughly **3.28 minutes** (down from historical averages of 4:00+). Many top hits, such as "i like the way you kiss me" or "Gata Only," are closer to the 2-3 minute mark.⁶
- **Economic Driver:** This is a direct result of streaming economics. Spotify pays per stream (provided it lasts 30 seconds). A 2:30 song can be streamed 24 times an hour, while a 4:00 song can only be streamed 15 times. Shorter songs maximize revenue potential and replay value, a trend visually confirmed by the dashboard's duration distribution.

7.4 Content: The Explicit Norm

The `` measure provides a cultural insight.

- **Data Point:** Approximately **35-40%** of the songs in the Top 50 are marked as "Explicit" [Image 2].
- **Insight:** Explicit content is no longer a barrier to commercial success. Driven by the dominance of Hip-Hop (Drake, Travis Scott, 21 Savage) and the raw authenticity favored by Gen Z listeners ⁷, explicit tracks perform exceptionally well. The "Explicit" tag, once a warning label, is now correlated with high streaming numbers in the Rap and Modern Pop genres.

7.5 Global Genre Diversification

The dataset highlights the decline of English-language hegemony.

- **Data Point:** Artists like **Bad Bunny** (Puerto Rico), **Peso Pluma** (Mexico), and **Jung Kook** (South Korea) are top-tier performers in the Global Top 50.¹⁵
 - **Insight:** The "Global" chart is truly globalizing. Latin music (Reggaeton/Regional Mexican) and K-Pop have mobilized fanbases that rival or exceed those of Western pop stars. This is visible in the *Popularity by Artist* chart, where non-English speaking artists consistently rank in the top 5 alongside Taylor Swift.
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8. CHALLENGES & LIMITATIONS

The development of the **Spotify Global Top 50 Dashboard** was not without challenges. Understanding these limitations is crucial for accurately interpreting the insights.

8.1 Data Granularity and "Streams" Absence

The provided dataset (spotify-top-50-world.csv) contains the popularity index (0-100) but lacks the raw **stream count** (e.g., "1,500,000 streams").

- **Limitation:** We cannot calculate exact revenue or market share percentages. We can only measure relative "trendiness." A song with a popularity of 95 is "hotter" than one with 90, but we don't know if that means 10% more streams or 50% more streams.²
- **Impact:** The dashboard focuses on *ranking* and *duration on chart* rather than financial volume.

8.2 Artist Data Normalization

The artist column often contains combined strings like "Future & Metro Boomin" or "Post Malone feat. Morgan Wallen."

- **Challenge:** In a simple aggregation, "Post Malone" and "Post Malone feat. Morgan Wallen" are treated as two different artists.
- **Mitigation:** While basic text cleaning was applied, a robust "Many-to-Many" relationship model would be required to perfectly attribute streams to every collaborating artist. In this dashboard, primary artists get the attribution, which slightly undercounts metrics for featured artists.

8.3 Volatility of the "Popularity" Algorithm

Spotify's popularity metric is dynamic and heavily weighted toward *recent* plays.

- **Challenge:** A classic song might have 1 billion total streams but a low popularity score today because it isn't currently trending.
- **Interpretation:** Users must understand that the dashboard measures *current cultural relevance* (2023-2024 trends), not historical legacy.

9. CONCLUSION & FUTURE SCOPE

9.1 Conclusion

The **Spotify Global Top 50 Dashboard** successfully transforms raw CSV data into a strategic asset for music industry analysis. The project demonstrates that the modern music landscape is defined by **high-velocity consumption** (favoring shorter singles), **superstar**

consolidation (Taylor Swift and Bad Bunny), and **global diversification** (Latin and K-Pop).

The analysis confirms that the "Single" is the primary tactical unit of the music business, used to maintain algorithmic momentum, while the "Album" remains the strategic unit for superstars to dominate market share. The visualization of these trends through Power BI provides stakeholders with the ability to monitor the pulse of global culture in real-time. The dashboard proves that data analytics is no longer just a support function in the music industry; it is the central nervous system of A&R and marketing strategy.

9.2 Future Scope

To elevate this project further, several enhancements are proposed:

- **Audio Features Integration:** Integrating Spotify's audio analysis API (Danceability, Energy, Tempo, Valence) would allow for sonic analysis. We could answer questions like "Do higher BPM songs chart better in the summer?"
- **Sentiment Analysis:** Applying Natural Language Processing (NLP) to song lyrics could categorize tracks by mood (Sad, Happy, Angry) to see how emotional sentiment correlates with global events or seasons.
- **Predictive Modeling:** Using historical data to build a Machine Learning model (e.g., Random Forest) to predict a song's future popularity score based on its initial 24-hour performance.
- **Real-Time API:** Transitioning from a static CSV to a live API connection would transform the dashboard into a real-time monitoring center for immediate decision-making.

(Student Signature)

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